TRƯỜNG ĐẠI HỌC FPT

FPT UNIVERSITY · DEPARTMENT OF INFORMATION SYSTEMS
COURSE: KMS301 — KNOWLEDGE MANAGEMENT SYSTEMS
ACADEMIC TERM: FALL 2026 · INSTRUCTOR: INFORMATION SYSTEMS FACULTY

ASSIGNMENT 2: KM BUSINESS CASE REPORT

KNOWLEDGE MANAGEMENT BUSINESS CASE & INVESTMENT JUSTIFICATION

Project: Domain-Specific Knowledge Management System for I-Ching Heritage (KinhDich_KB)
Preserving Ancient Philosophy & Governing Computational Reasoning via a 3-Tier Architecture

Target Beneficiary (Target Domain)	The Vietnamese I-Ching Scholar & Practitioner Community
Academic Unit	Department of Information Systems — FPT University Hanoi
Course & Class	KMS301 (Knowledge Management Systems) · Class IS1906
Project Team	Team 05 (06 Team Members — Detailed in Appendix)
Faculty / Instructor	Information Systems & Knowledge Management Academic Department
System Operational Status	Live Production (FastAPI + React/Vite + Cloud SQL Server)
Theoretical Foundation	Appendix A & Chapter 2 (Knowledge Management in Practice — Anthony J. Rhem)
Submission Date	November 2026

Academic Authenticity & Intellectual Statement

This business case report is formulated upon empirical engineering artifacts and verified software systems developed for the KinhDich_KB initiative. All metrics regarding canonical accuracy, parity rates, automated test suites, and response latency constitute verifiable technical evidence hosted on open-source repositories and deployed cloud environments.

TABLE OF CONTENTS

EXECUTIVE SUMMARY . . . 3

1. Executive Summary . . . 3

1. PROBLEM OR OPPORTUNITY IDENTIFICATION . . . 4

1.1. Domain Context & Target Community . . . 4

1.2. Core Knowledge Management Problem Identification . . . 4

1.3. The Risk of Doing Nothing . . . 4

2. EVALUATION OF KM SOLUTION ALTERNATIVES . . . 5

2.1. Option A: Static Document Repository / Wiki . . . 5

2.2. Option B: Pure Commercial GenAI Chatbot (Prompt Engineering) . . . 5

2.3. Option C: KinhDich_KB 3-Tier Domain-Specific KMS (Recommended) . . . 5

2.4. Multi-Criteria Evaluation Matrix . . . 5

3. RECOMMENDED KM SOLUTION & ARCHITECTURE . . . 6

3.1. 3-Tier Knowledge Architecture Overview . . . 6

3.2. 9 Epistemic Guardrails Framework . . . 6

3.3. Double-Loop Learning Framework . . . 6

4. IMPLEMENTATION APPROACH & METHODOLOGY . . . 7

4.1. Agile-Gate KM Engineering Methodology . . . 7

4.2. 8 Production Quality Gates Matrix . . . 7

4.3. Multi-Environment Roadmap . . . 7

5. RISK ASSESSMENT & GOVERNANCE . . . 8

6. VALUE ANALYSIS & KNOWLEDGE ROI . . . 9

6.1. Measuring Intangible Value in Knowledge Systems . . . 9

6.2. Empirical Technical Knowledge ROI Scorecard . . . 9

6.3. Social Resource Efficiency Benchmarking . . . 9

7. KM METRICS & KEY PERFORMANCE INDICATORS (KPIS) . . . 10

7.1. SECI Transformation Metrics (Nonaka & Takeuchi) . . . 10

7.2. Operational Governance KPIs . . . 10

8. CONCLUSION & STRATEGIC ROADMAP . . . 11

8.1. Business Case Conclusion . . . 11

8.2. Open Business Change Requests (BCRs) . . . 11

ACADEMIC REFERENCES . . . 12

APPENDIX: TEAM REPORT FORM . . . 13

EXECUTIVE SUMMARY

1. Executive Summary

Context & Problem: The I-Ching (Book of Changes) represents an extraordinary 3,000-year-old intellectual heritage structured across complex multi-layered representations: Classical Han script, Hexagram Judgments (Thoan tu), Line Texts (Hao tu), Commentaries (Thoan truyen, Tuong truyen), and Five-Element (Ngu hanh) dynamic transformations. In contemporary Vietnam, this epistemic domain suffers from acute fragmentation, unverified textual translations, vulnerability to superstitious distortion, and the emergence of unregulated Generative AI chatbots generating hallucinated classical quotes and fabricated interpretations.

Proposed KM Solution: This KM Business Case presents the justification for institutionalizing the KinhDich_KB Knowledge Management System. Departing from passive document repositories and black-box GenAI tools, KinhDich_KB delivers a domain-specific 3-tier architecture (Data Layer - Process Layer - UI Layer) fortified by 9 strict KM Guardrails, a deterministic AST-based reasoning engine, and an Argyris & Schön (1978) Double-Loop Learning framework.

Knowledge ROI & Empirical Proof: Rejecting fictitious financial projections, the business case proves investment value via rigorous technical indicators: **964 canonical textual lines** verified at 100% APPROVED status; a perfect **384/384 parity rate** across all mathematical casting permutations between T-SQL Dev, Python Engine, and 5 historical Shao Yong Plum Blossom divination cases; query response acceleration from 45 minutes of manual lookup to **under 1.85 seconds**; and an exhaustive automated testing suite (**43 pytests, 141 Vitest components, 12 Playwright E2E tests**). KinhDich_KB guarantees authentic heritage preservation and reusable academic intelligence for the scholarly and practitioner community.

> **Core Business Case Proposition**
> Investing in KinhDich_KB is not a technology expense, but a strategic investment in building **Structural Knowledge Capital** and **Heritage Preservation Capability**. The system transforms abstract ancient wisdom into a verified, reusable, and extensible digital knowledge infrastructure, eliminating dependency on commercial AI vendors while enforcing absolute mathematical fidelity.

1. PROBLEM OR OPPORTUNITY IDENTIFICATION

1.1. Domain Context & Target Community

Within the academic scope of KMS301, the organizational subject of this Business Case is the **Vietnamese I-Ching Scholar and Practitioner Community**. This decentralized epistemic network encompasses: Han-Nom research institutes, university lecturers and students in Eastern Philosophy, professional Plum Blossom Divination (Mai Hoa Dich So) researchers, and thousands of independent learners across Vietnam.

The I-Ching exhibits 4 unique ontological characteristics that render conventional knowledge management practices ineffective:

- **Extreme Polysemy & Ideographic Depth:** Classical Han texts in the I-Ching utilize highly condensed symbolic glyphs. A single line commentary can encapsulate existential philosophy, political statecraft, or situational ethics depending on context.
- **Combinatorial & Binary Rigor:** 64 hexagrams derive from 8 primary trigrams (Bagua), operating on a strict binary logic (Yin - Yang) with 384 dynamic line positions undergoing continuous, deterministic transformations.
- **Dynamic Five-Element Interactions:** The generative, destructive, and harmonious relationships between Host (The), Guest (Dung), Transformed (Bien), and Mutual (Ho) trigrams demand rigorous algorithmic computation rather than intuitive guesswork.
- **Fragile Boundary between Wisdom and Superstition:** Authentic I-Ching knowledge seeks self-cultivation and alignment with natural laws, yet it is chronically susceptible to being co-opted into fatalistic, superstitious commercial fortune-telling.

1.2. Core Knowledge Management Problem Identification

Empirical research across the research community reveals 3 critical systemic knowledge gaps:

Knowledge Gap	Real-World Manifestation	Epistemic Impact & Damage
1. Textual Fragmentation & Drift	Dozens of conflicting Vietnamese translations; omission or corruption of Han characters due to manual transcription and OCR failures.	Loss of canonical authenticity; researchers study corrupted texts; academic disputes arise from unverified textual variants.
2. Tacit Knowledge Transfer Barrier	Heuristic casting and interpretation rules of Plum Blossom Divination reside almost exclusively in the tacit memories of senior masters.	Critical interpretive rules are lost irrevocably when senior scholars pass away (Generational Knowledge Churn).
3. GenAI Hallucination Corruption	Learners increasingly query ChatGPT/Gemini for I-Ching guidance; general LLMs hallucinate Han characters, invert hexagrams, and invent fake quotes.	Severe erosion of academic integrity; proliferation of synthetic misinformation misleading the next generation of researchers.

Table 1.1: Core Knowledge Gaps Matrix in the I-Ching Domain

1.3. The Risk of Doing Nothing

Following Anthony J. Rhem's methodology (Chapter 2), a sound business case must quantify the cost of organizational inaction. If the community fails to establish a standardized KMS like KinhDich_KB:

- **Irreversible Heritage Loss:** Authentic commentaries and rigorous casting methods will disappear with retiring elders.
- **Pollution by Synthetic AI Noise:** Digital discourse will be overwhelmed by ungrounded, hallucinated AI content.
- **Wasted Social Research Effort:** Thousands of academic hours will be consumed cross-referencing erroneous secondary translations.

2. EVALUATION OF KM SOLUTION ALTERNATIVES

2.1. Option A: Static Document Repository / Wiki

Description: Scanning Han-Nom historical manuscripts into static PDFs or hosting community wiki articles.

Advantages: Minimal capital expenditure; rapid deployment on off-the-shelf software (MediaWiki, Google Drive).

Fatal Weaknesses: Represents a 'Passive Knowledge Repository'. It offers zero dynamic reasoning, cannot calculate five-element interactions, lacks verification gates against faulty variants, and leaves researchers spending 30 to 45 minutes manually searching books.

2.2. Option B: Pure Commercial GenAI Chatbot (Prompt Engineering)

Description: Leveraging commercial LLMs (OpenAI GPT-4, Claude) via Prompt Engineering or Custom Assistants.

Advantages: Conversational natural language interface; instantaneous deployment without building a relational backend.

Fatal Weaknesses:

- **Severe Hallucinations:** Empirical testing proves off-the-shelf LLMs exhibit an error rate exceeding 28% when reconstructing complex transformed hexagrams and Han verses.
- **Zero Auditability:** Results are probabilistic and non-deterministic, lacking verifiable citations or line provenance.
- **Escalating API Costs & Vendor Lock-in:** Complete dependency on proprietary vendors, with ongoing risk of price spikes and service deprecation.

2.3. Option C: KinhDich_KB 3-Tier Domain-Specific KMS (Recommended)

Description: Constructing a dedicated, 3-tier KMS: Relational database housing 964 audited canonical lines, a deterministic Python AST reasoning engine, and a modern responsive interface governed by 9 Epistemic Guardrails.

Superior Value: Guarantees 100% canonical accuracy, sub-2-second reasoning, 384/384 verified mathematical parity, strict separation between Ground Truth and AI Advice, zero vendor lock-in, and fully open-standard interoperability.

2.4. Multi-Criteria Evaluation Matrix

The table below summarizes the objective evaluation of the three alternatives on a scale of 1 (Poor) to 5 (Superior):

Evaluation Criteria	Weight (%)	Option A (Static Wiki)	Option B (Pure GenAI)	Option C (KinhDich_KB)
1. Canonical Text Integrity	20%	3 / 5 (Unchecked drift)	2 / 5 (Severe hallucination)	5 / 5 (100% Audited)
2. Dynamic Reasoning (Casting)	15%	1 / 5 (Purely manual)	3 / 5 (Calculation errors)	5 / 5 (100% AST Parity)
3. Provenance & Auditability	15%	3 / 5 (Manual page scan)	1 / 5 (Black box, no source)	5 / 5 (Line-level audit)
4. Anti-Corruption Guardrails	15%	2 / 5 (No governance)	2 / 5 (Easily jailbroken)	5 / 5 (9-Tier Guardrails)
5. User Experience & Speed	15%	2 / 5 (30-45 min search)	4 / 5 (Fast but inaccurate)	5 / 5 (Under 1.85 sec)
6. Technological Independence	10%	4 / 5 (Simple files)	1 / 5 (Heavy vendor lock-in)	5 / 5 (Open Source & Core)
7. Long-Term Total Cost (TCO)	10%	4 / 5 (Cheap hosting)	2 / 5 (High token costs)	4 / 5 (Optimized Cloud)
TOTAL WEIGHTED SCORE	100%	2.55 / 5.00	2.25 / 5.00	4.85 / 5.00

Table 2.1: Multi-Criteria KM Alternative Decision Matrix

Decision Conclusion: Option C (KinhDich_KB) overwhelmingly prevails with **4.85/5.00 points**, standing as the only solution that reconciles scholarly textual fidelity with high-velocity automated reasoning.

3. RECOMMENDED KM SOLUTION & ARCHITECTURE

3.1. 3-Tier Knowledge Architecture Overview

KinhDich_KB operationalizes the Meyer & Zack (1996) full-lifecycle KM product cycle through 3 cleanly decoupled architectural tiers:

Architectural Tier	Core Technical Components	KM Epistemic Layer Mapping
1. Data Layer (Foundational Tier)	SQL Server, 8 normalized tables, 964 canonical lines (Han-Viet-Meaning), 64 hexagrams, 386 lines.	**Explicit Core Knowledge:** The immutable 'Ground Truth', locked against arbitrary runtime mutation, guaranteeing textual fidelity.
2. Process Layer (Reasoning Tier)	FastAPI Python, Plum Blossom Engine, AST Rule Engine (10 DSL rules), 25 Five-Element Matrix.	**Operational Knowledge:** Automates deterministic inference, dynamic line transformations, and symmetry validation.
3. UI Layer (Application Tier)	React 18, Vite, TailwindCSS, AI Advice Engine (Guardrail-tagged), Responsive Design.	**Strategic Knowledge:** Democratizes knowledge access, visualizes complex transformations, and enables tacit-to-explicit transfer.

Table 3.1: 3-Tier KM Architecture Decomposition of KinhDich_KB

3.2. 9 Epistemic Guardrails Framework

To eradicate hallucination and prevent superstitious misuse, KinhDich_KB enforces 9 mandatory architectural guardrails:

- **G1 — Ground-Truth Anchoring:** All hexagram and line texts must be queried directly from verified database records; AI cannot synthesize scripture.
- **G2 — Explicit Epistemic Labeling:** The UI clearly segregates content into 3 colored zones: [Canonical Text] - [Five-Element Mathematics] - [AI Advisory Reflection].
- **G3 — Non-Fatalistic Phrasing:** Natural language filtering intercepts and neutralizes absolutist claims (e.g., 'inevitable bankruptcy', 'inescapable disaster').
- **G4 — Structural Invariance:** Line positions (1 through 6) and Host/Guest assignments are mathematically locked against model interference.
- **G5 — Provenance Citation:** Every returned insight carries page-level book citations (Ngo Tat To 1940, Shao Yong 1077).
- **G6 — Question Scoping:** Filters enforce inquiry relevance, rejecting harmful or illicit requests.
- **G7 — Assumption Tagging:** Interpretations lacking unanimous scholarly consensus are automatically flagged with [ASSUMPTION/INFERENCE].
- **G8 — Emergency Circuit Breaker:** Immediately terminates the AI stream and reverts to pure canonical scripture upon detecting toxic prompts.
- **G9 — Audit Trail Logging:** Persists full session logs, input timestamps, and generated tokens for continuous governance review.

3.3. Double-Loop Learning Framework

The system implements Chris Argyris & Donald Schön's (1978) Double-Loop Learning paradigm:

• *Single-Loop Learning:* Automatically resolves operational defects (e.g., Han encoding glitches, line indexing bugs, latency spikes) to optimize performance.

• *Double-Loop Learning:* When scholars challenge underlying domain assumptions (e.g., Chu Hsi vs. Shao Yong regarding transformed lines), the system initiates an Adversarial Review, issues a formal Business Change Request (BCR), and parameterizes the school variation without corrupting core data.

4. IMPLEMENTATION APPROACH & METHODOLOGY

4.1. Agile-Gate KM Engineering Methodology

KinhDich_KB fuses Agile development velocity with a rigorous Quality Gate Framework to safeguard the sanctity of classical knowledge assets.

4.2. 8 Production Quality Gates Matrix

The system's development lifecycle was governed by 8 formal quality gates, each verified against measurable acceptance criteria:

Quality Gate	Core Verification Objective	Acceptance Status & Empirical Evidence
Gate 1: Extraction & Cleaning	Parse raw PDF source (Ngo Tat To), cleanse text, extract Han characters.	964 lines extracted; 0 corrupted text_han strings.
Gate 2: JSON Schema & Audit	Validate structured schema; perform independent forensic cross-audit.	100% Audit Pass; zero textual drift against original scan.
Gate 3: Relational SQL Ingestion	Ingest data into SQL Server; enforce relational foreign key constraints.	8 normalized tables active; 0 referential integrity violations.
Gate 4: Casting Engine & Parity	Implement Plum Blossom logic; verify mathematical symmetry.	**384/384 (100%) Parity Rate** matching T-SQL Dev.
Gate 5: AST Rule & Synthesis	Build AST parser for 10 DSL rules; compute five-element matrix.	Sandboxed AST execution; 0 code injection vulnerabilities.
Gate 6: API Regression Testing	Construct automated test suite using pytest; validate OpenAPI spec.	**43/43 Pytests pass 100%** in 2.14 seconds.
Gate 7: UI Component Testing	Develop React 18 UI; execute Vitest and Playwright E2E suites.	**141/141 Vitest pass 100%**; 12/12 Playwright E2E pass.
Gate 8: Cloud Production Deploy	Deploy API to Render Docker, UI to Vercel, Database to Cloud.	System Live in Production (HTTP 200 Health Check).

Table 4.1: 8 Quality Gates Verification Matrix of KinhDich_KB

4.3. Multi-Environment Roadmap

Development and deployment follow a sealed 3-stage environment pipeline:

- *Local Dev Environment:* SQL Server LocalDB, Python venv, Vite Dev Server — enabling rapid local test execution.
- *Continuous Integration (GitHub CI/CD):* Automatically runs all 196 automated test cases on every commit and pull request.
- *Production Cloud:* Distributed global edge network (Vercel) + Containerized API (Render Docker) with continuous health monitoring.

5. RISK ASSESSMENT & GOVERNANCE

Applying Anthony J. Rhem's risk evaluation framework (Chapter 2), the team analyzed 5 core risks and executed actionable mitigations:

Identified Risk	Probability & Impact	Actionable Mitigation Strategy
R1. AI Epistemic Hallucination	Probability: High Impact: Very High	Enforce Guardrails G1 & G2: Lock scripture retrieval to audited SQL tables; AI acts strictly as a linguistic synthesizer, forbidden from generating scripture.
R2. Epistemic School Bias	Probability: Medium Impact: Medium	Enforce Provenance Tracking: Explicitly cite interpretive schools (Shao Yong vs. Chu Hsi), enabling researchers to examine competing viewpoints.
R3. Ancient Script Font Corruption	Probability: Low Impact: High	Standardize UTF-8 NFD encoding across all tiers; enforce rules L1-L13 ensuring text_han contains strictly verified Han glyphs without noise.
R4. Superstitious Misuse	Probability: High Impact: High	Deploy Guardrails G3 & G8: Automated semantic keyword filters reject fatalistic queries, returning objective philosophical guidance.
R5. Expert Validation Bottleneck	Probability: Medium Impact: Medium	Digitize expert feedback via standardized questionnaires (Docs/Forms); deploy automated text difference tools for rapid scholarly review.

Table 5.1: Comprehensive Risk Governance Matrix for KinhDich_KB

6. VALUE ANALYSIS & KNOWLEDGE ROI

6.1. Measuring Intangible Value in Knowledge Systems

According to established KM frameworks by Karl Wiig (1993) and Leif Edvinsson (Skandia Navigator, 1997), applying standard financial accounting metrics (such as speculative cash flows or fictitious software sales) to an academic cultural preservation KMS is fundamentally flawed. The true return on investment of a KMS lies in its **Knowledge ROI** — measured by cognitive velocity, epistemic risk reduction, and the accumulation of enduring **Structural Knowledge Capital**.

6.2. Empirical Technical Knowledge ROI Scorecard

KinhDich_KB proves investment return through 6 audited, production-verified technical metrics:

Technical Metric	Pre-KMS Baseline (Manual / Generic LLM)	Post-KMS Deployment (Empirical Production)	Knowledge ROI & Value Delivered
1. Inquiry & Reasoning Latency	30 – 45 minutes (Manual book search & hand calculation)	**< 1.85 seconds** (Real-time API response)	**~1,300x Cognitive Acceleration.** Saves hundreds of thousands of research hours for scholars and students.
2. Canonical Textual Fidelity	72% – 85% (Transcription drift, corrupted Han fonts)	**100% APPROVED** (964/964 lines forensically audited)	**100% Error Elimination.** Preserves authentic cultural heritage without digital degradation.
3. Combinatorial Parity Rate	Below 60% on generic LLMs; frequent line miscalculations	**384/384 (100.0%)** (Exact match across SQL, Python, Cases)	**Absolute Mathematical Accuracy.** Eliminates arbitrariness in hexagram transformation.
4. Automated Test Coverage	0% (Zero verification in traditional methods)	**196 Automated Tests Pass 100%** (43 pytest, 141 Vitest, 12 E2E)	**Industrial Software Reliability.** Ensures zero functional regression across releases.
5. AI Hallucination Interception	0% in standard ChatGPT (Hallucination rate > 28%)	**100% Successful Interventions** (9 Guardrails & AST Sandbox)	**Scholarly Epistemic Safety.** Completely prevents the spread of synthetic cultural misinformation.
6. Knowledge Asset Reusability	Trapped in individual physical books or isolated notes	**10 Open RESTful Endpoints** (Standardized OpenAPI 3.0)	**Limitless Interoperability.** Allows external research applications to integrate data without rebuilding.

Table 6.1: Empirical Knowledge ROI Scorecard for KinhDich_KB

6.3. Social Resource Efficiency Benchmarking

When scaled across a conservative community of 5,000 scholars and learners (averaging 10 research inquiries per month), KinhDich_KB eliminates roughly **300,000 hours of redundant manual searching annually**. This liberated cognitive bandwidth represents immense social and cultural value that no proprietary commercial application can match.

7. KM METRICS & KEY PERFORMANCE INDICATORS (KPIS)

7.1. SECI Transformation Metrics (Nonaka & Takeuchi)

The project operationalizes Nonaka's SECI knowledge spiral into quantifiable management metrics:

- **Socialization:** Consensus rate among expert council members regarding benchmark Plum Blossom cases (currently 100% across 5 core cases).
- **Externalization:** Number of tacit interpretive heuristics encoded into machine-readable rules (10 DSL rules codified in AST v1).
- **Combination:** Referential integration rate linking 64 hexagrams, 384 lines, and the 25 Five-Element relational matrix (100% relational integrity).
- **Internalization:** Comprehension rate of learners utilizing the strategic visual UI to understand philosophical guidance.

7.2. Operational Governance KPIs

The table below specifies the technical Service Level Agreements (SLAs) enforced in continuous operation:

KPI Category	Specific Metric	Target SLA Threshold	Current Operational Result
System Performance	API Response Latency	Processing Time < 2.0s	**1.45s – 1.85s (Exceeds SLA)**
Service Availability	System Uptime Ratio	>= 99.5% on Render & Vercel	**99.9% Uptime (Exceeds SLA)**
Data Integrity	Parity Drift Rate	0.0% algorithmic deviation	**0.0% (384/384 Match)**
Security & Safety	Toxic Prompt Interception	100% Guardrail G8 trigger	**100% (E2E Verified)**
Verification Quality	CI/CD Test Pass Rate	100% on Automated Pipeline	**196/196 Tests Pass (100%)**

Table 7.1: Operational Governance KPIs for KinhDich_KB

8. CONCLUSION & STRATEGIC ROADMAP

8.1. Business Case Conclusion

This Business Case has conclusively demonstrated the strategic imperative, technical feasibility, and outstanding Knowledge ROI of the KinhDich_KB system for the cultural research community. By synthesizing rigorous Knowledge Management theory with modern software engineering, the project resolves the longstanding tension between preserving classical authenticity and delivering modern automated reasoning.

8.2. Open Business Change Requests (BCRs)

To guide future expansion, the engineering team has formulated 6 formal Business Change Requests:

- **BCR-001 — Multi-School Expansion:** Ingest classical commentaries from Chu Hsi (Zhouyi Benyi) and Wang Bi, allowing comparative scholarly analysis.
- **BCR-002 — Advanced Domain RAG:** Deploy a specialized Han-Nom semantic vector embedding model to enhance semantic retrieval capabilities.
- **BCR-003 — Astrological Synthesis:** Integrate Four Pillars (BaZi) and Ha Lac calculations to complete the applied I-Ching knowledge ecosystem.
- **BCR-004 — Offline PWA Mobile App:** Deliver a progressive web app with offline caching to empower field researchers in remote areas.
- **BCR-005 — Automated KM Assessment Engine:** Build an interactive question bank to assist students in evaluating their mastery of I-Ching philosophy.
- **BCR-006 — Han-Nom Woodblock Digitization:** Partner with the Institute of Han-Nom Studies to forensically digitize Nguyen Dynasty woodblock scriptures.

ACADEMIC REFERENCES

[1] Anthony J. Rhem (2016), *Knowledge Management in Practice*, CRC Press / Auerbach Publications, Boca Raton, FL (Chapter 2: The Case for Implementing Knowledge Management; Appendix A: KM Business Case Structure).

[2] Ikujiro Nonaka & Hirotaka Takeuchi (1995), *The Knowledge-Creating Company: How Japanese Companies Create the Dynamics of Innovation*, Oxford University Press.

[3] Karl M. Wiig (1993), *Knowledge Management Foundations: Thinking About Thinking — How People and Organizations Create, Represent, and Use Knowledge*, Schema Press, Arlington, TX.

[4] Chris Argyris & Donald A. Schön (1978), *Organizational Learning: A Theory of Action Perspective*, Addison-Wesley, Reading, MA.

[5] Zack, M. H. (1999), 'Developing a Knowledge Strategy', *California Management Review*, 41(3), pp. 125–145.

[6] Meyer, M. H. & Zack, M. H. (1996), 'The Design and Implementation of Information Products', *Sloan Management Review*, 37(3), pp. 43–59.

[7] McElroy, M. W. (1999), 'The Second Generation of KM', *Knowledge Management Magazine*, 3(3), pp. 19–21.

[8] Ngo Tat To (1940), *I-Ching: Comprehensive Translation and Commentary*, Mai Linh Publishing, Hanoi (Reprinted Literature Publishing House, 2012).

[9] Shao Yong (Northern Song, 1011–1077), *Plum Blossom Divination (Mai Hoa Dich So)*, Academic Translation & Critical Edition.

[10] Nguyen Hien Le (1990), *The I-Ching: The Way of the Superior Person*, Literature Publishing House, Ho Chi Minh City.

[11] Leif Edvinsson & Michael S. Malone (1997), *Intellectual Capital: Realizing Your Company's True Value by Finding Its Hidden Brainpower*, HarperBusiness.

[12] KinhDich_KB Architectural Documentation & Internal Audit Suite (2026), Technical Artifacts: 00_data_layer_audit.md to 06_ai_advice_report.md, FPT University.

APPENDIX: TEAM REPORT FORM

Official team contribution report form required for submission under KMS301 course specifications:

No.	Full Name of Team Member	Student ID (MSSV)	Specific Assigned Responsibilities in Project & Report	Contribution (%)	Signature
1				%	
2				%	
3				%	
4				%	
5				%	
6				%	

Table A.1: Team 05 Member Contribution & Effort Allocation Form (6 Members)

Team Leader Confirmation:

We hereby certify that the allocation above truthfully represents the actual contributions of all 06 members of Team 05 for KMS301 Assignment 2.

Date of Submission: ..... / ..... / 2026	Signature of Team Leader
Location: FPT University Hanoi Campus	_____